PROJECT DOCUMENTATION

Data Analysis for Environment and Sustainability

Environmental data integration, exploratory analysis, predictive modeling, and clustering report

Field	Details
Supervisors	Prof. Noha Gamal Eldin Saad; Eng. Lamiaa Omar Hassan
Study Period	Daily city-level observations for 2023
Cities	Cairo, Dubai, London, New York, Tokyo, Paris, Nairobi
Deliverable	Main project documentation report

Project Members

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Executive Summary

This project builds an integrated urban environmental dataset to study how air pollution, green space, weather, electricity consumption, and day type interact across seven major cities. The final dataset contains daily 2023 observations after collection, cleaning, merging, preprocessing, analysis, modeling, and application development.

The main analytical story is that urban environmental quality is not explained by one factor alone. The analysis shows that green space is associated with lower pollution, weekends are slightly cleaner and more stable, monthly electricity use changes alongside temperature fluctuations, and cluster modeling reveals structural city profiles that persist across seasons.

Project Fact	Value
Final dataset	2,548 rows and 17 documented columns
Coverage	2023-01-01 to 2023-12-31
Cities	Cairo, Dubai, London, Nairobi, New York, Paris, Tokyo
Missing values after preprocessing	0
Core research questions	Green space vs air quality; weekend vs weekday air quality; temperature vs electricity consumption
Modeling outputs	Random Forest PM2.5 predictor and K-Means humidity cluster predictor
Applications	Streamlit dashboard for PM2.5 prediction and humidity cluster prediction

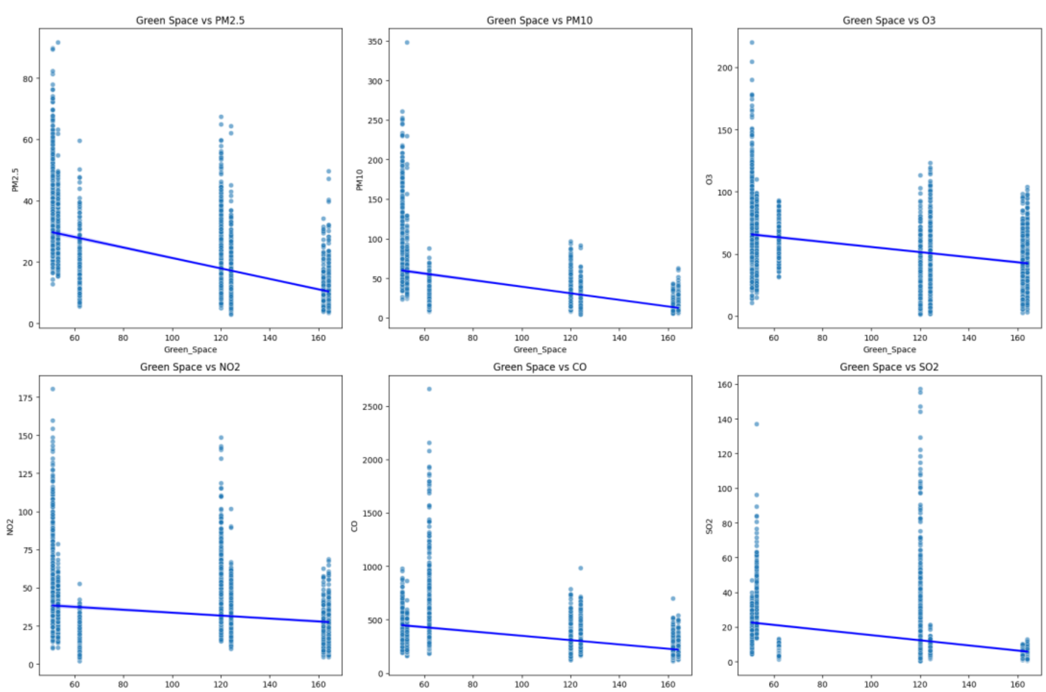


Figure 1. Green-space vs pollutant scatter plots.

Why We Chose This Project

We chose environment and sustainability because cities are where climate pressure, population density, public health, energy demand, and policy choices meet. Air quality is directly connected to human health, while electricity demand and green space reflect how urban systems either intensify or reduce environmental stress.

The project also gives a strong data-analysis story: weather conditions influence energy demand, electricity demand can reflect human activity and emissions pressure, emissions influence air quality, and green space can help mitigate pollution. This makes the topic suitable for integration, visualization, statistical analysis, supervised modeling, and clustering.

Why These Seven Cities

City Group	Cities	Reason for Inclusion
Hot arid urban systems	Cairo and Dubai	High temperatures and low green-space values create strong contrast for pollution and electricity-demand analysis.
Temperate global cities	London, Paris, New York, Tokyo	Large international cities with different policy, density, climate, and green-space profiles.
Equatorial/highland reference	Nairobi	Adds a different African climate profile and reveals patterns not explained only by city size or income.
Overall selection logic	All seven	Together they provide geographic diversity, climate contrast, urban-scale variation, and enough data availability for a consistent 2023 comparison.

Why These Features

The features were selected to represent four connected systems: pollution exposure, weather conditions, urban mitigation, and human or infrastructure demand. Pollutants capture air-quality risk; weather features explain dispersion and seasonal behavior; green space represents urban environmental protection; electricity consumption approximates energy demand and activity pressure.

Data Collection

The project combines public APIs, source-specific extraction, and project-level generated estimates. Every source was transformed into a daily city-level structure so that records could be compared and merged consistently.

Dataset	Collection Method	Source	Key Fields / Notes
Air quality	Hourly API records aggregated into daily averages	Open-Meteo air-quality API	PM2.5, PM10, O3, NO2, CO, SO2 for each city and date.
Weather	Historical observations aggregated to daily values	Weather Underground / weather.com	Temperature mean/max, humidity, wind speed, pressure.
Green space	Scraped static city-level green-space values	StatRanker green-space ranking	Assumed stable across 2023 because green-space availability changes slowly.
Electricity	Monthly values expanded across days for merging	EIA API for New York; estimated monthly profiles for other cities	City, Date, Electricity Consumption; repeated daily within each month.

How the Data Was Merged

First, the air-quality data was merged with green-space data on City. Green space is static, so each city value was repeated for all dates of that city. Second, each city's weather file was cleaned and concatenated into one weather table. Third, the air-quality/green-space table, the weather table, and the electricity table were merged using City and Date as the common keys.

After merging, two derived time features were added. The isWeekend flag uses Friday/Saturday for Cairo and Dubai and Saturday/Sunday for the other cities. The Season field maps months into Winter, Spring, Summer, and Autumn. The final dataset was then validated for duplicate rows and missing values.

City	Final Rows
Cairo	365
Dubai	365
London	365
Nairobi	363
New York	360
Paris	365
Tokyo	365

Feature Documentation

Category	Feature	Documentation
Location/time	City	Groups observations by urban area and enables cross-city comparisons.
Location/time	Date	Daily time key used for joining, trend analysis, weekends, and seasons.
Air pollution	PM2.5	Fine particles linked to health risk and used as the main prediction target.
Air pollution	PM10	Larger particulate matter; useful for dust and source patterns.
Air pollution	O3	Ground-level ozone; tends to respond to sunlight and warmer conditions.
Air pollution	NO2	Traffic and combustion-related pollutant.
Air pollution	CO	Combustion-related gas; important for episodic pollution events.
Air pollution	SO2	Fossil-fuel and industrial emission indicator.
Urban environment	Green_Space	Urban mitigation feature for the green-space research question.
Weather	Temperature_mean	Average daily temperature; linked to pollution chemistry and electricity demand.
Weather	Temperature_max	Daily peak temperature; useful for extreme heat analysis.
Weather	Humidity	Affects pollutant behavior, comfort, and atmospheric stability.
Weather	Wind_speed	Represents pollutant dispersion.
Weather	Pressure	Atmospheric pressure; partly influenced by city altitude.
Energy	Electricity Consumption	Monthly electricity signal repeated daily to align with city-date analysis.
Time behavior	isWeekend	Separates working-day and weekend activity patterns.
Time behavior	Season	Supports seasonal pollution and clustering analysis.

Main Data Issues and Handling

Issue	What Happened	Handling Decision	Why This Was Acceptable
New York missing weather values	Project notes identified missing January temperature and humidity values. Raw inspection showed an initial January block plus one April value missing for temperature mean, temperature max, and humidity.	Backfill was applied to New York weather values, filling missing records from the next available observation.	Backfill kept the New York series usable for merging and avoided deleting a major city from early-year analysis.
Electricity data monthly, not daily	Electricity consumption was available or estimated at monthly granularity.	Each day in a month received the same monthly value.	This preserved monthly energy pattern while enabling daily joins with weather and air quality.

Date alignment	Some weather files contained a 2022-12-31 row or ended before 2023-12-31.	Extra leading rows were removed where needed and inner joins were used on City and Date.	The final dataset contains only rows with complete joined evidence across all required tables.
Different feature scales	Pollution, weather, electricity, and green-space values use different units.	Scaling was applied before clustering and normalization was used for combined pollution indexes.	This prevented large-scale variables from dominating distance-based models and combined scores.

Validation result: the final file has zero duplicate rows and zero missing values after preprocessing.

Main Research Questions and Insights

Question 1: Does Air Quality Improve With More Green Space?

Several pollutants show negative relationships with green space. Cities with larger green areas often demonstrate lower pollution levels, and the combined air-quality score generally decreases as green space increases.

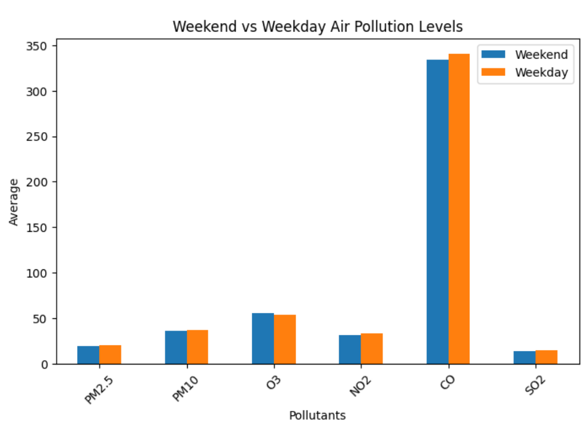
It also concludes that air pollutants tend to rise together in heavily polluted cities. Overall, increasing urban green space may contribute to healthier environmental conditions and reduced air-pollution exposure.

Question 2: Are Pollution Levels Lower on Weekends?

The average pollutant bar plot does not show a major difference between weekdays and weekends. After normalization, however, the boxplot indicates that pollution levels are generally lower and more stable on weekends, with a reduced median and shorter upper whisker compared with weekdays.

Both periods can experience sudden spikes, but weekdays show a denser cluster of extreme outliers. Overall, air quality improves on weekends when industrial activity and commuter traffic decline, while the bar chart reinforces that average pollution levels are slightly lower on weekends.

Day Type	PM2.5	PM10	O3	NO2	CO	SO2
Weekend	19.799664	35.861166	55.263053	31.744340	334.224521	14.069140
Weekday	20.666187	37.538924	54.240919	33.723271	340.303385	14.619999



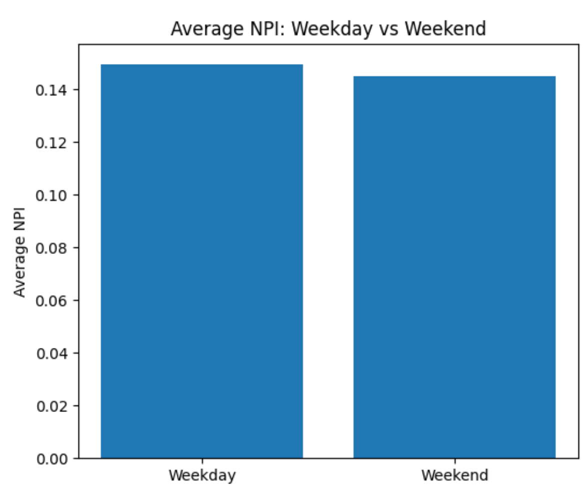


Figure 2. Weekend vs weekday pollutant averages.

Figure 3. Normalized pollution index by day type.

Question 3: How Does Temperature Relate to Electricity Consumption?

The observed Pearson correlation between temperature and electricity consumption is negative 0.18623927732177423. Electricity consumption changes across different months in all cities, and higher temperatures during warmer months are generally associated with higher electricity consumption because of increased cooling demand.

Temperature patterns differ between cities depending on climate conditions. The scatter plot and correlation analysis indicate a noticeable relationship between temperature and electricity consumption, seasonal changes play an important role in electricity usage patterns, and the monthly trend visualization shows electricity usage changing alongside temperature fluctuations over the year.

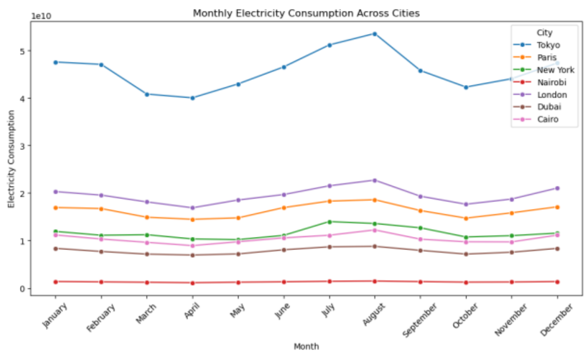


Figure 4. Monthly electricity consumption.

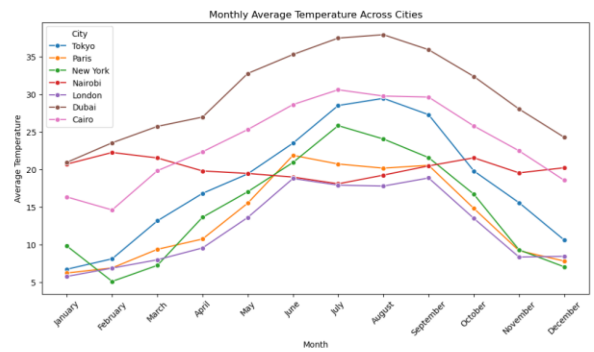


Figure 5. Monthly temperature changes.

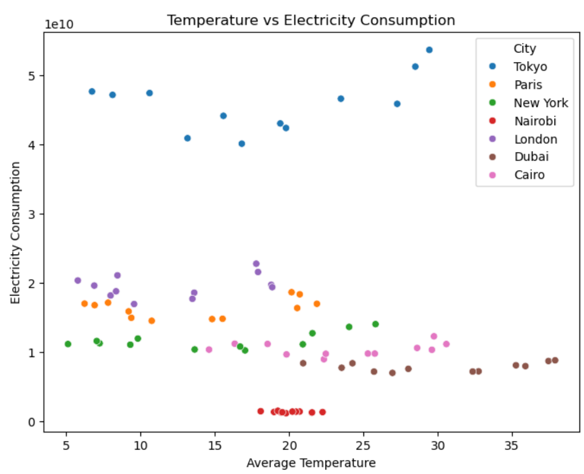


Figure 6. Temperature vs electricity consumption.

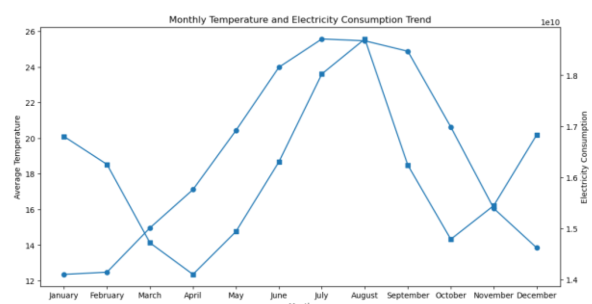


Figure 7. Monthly temperature and electricity trend.

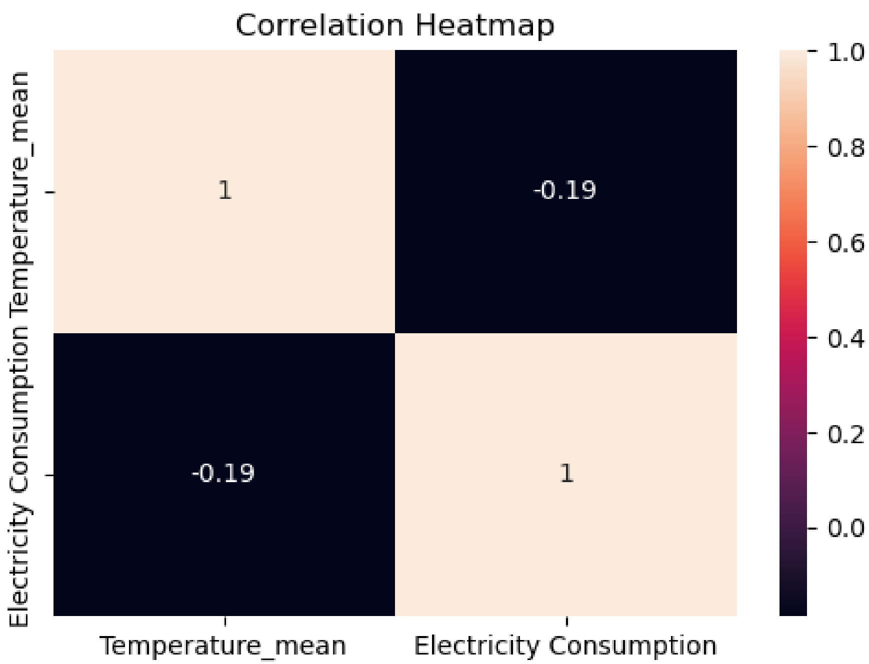


Figure 8. Temperature and electricity correlation heatmap.

Extra Environmental Questions

Extra Question	Main Insight	Interpretation
Which season has the worst overall air quality?	Summer has the highest normalized pollution index, followed closely by Spring. Autumn is the lowest.	Warm periods increase ozone and particulate-related pressure in several cities, but city-specific seasonality still matters.
Which weather conditions are linked with worse air quality?	Hotter and drier days are associated with worse pollution. Humidity and wind speed are generally linked with cleaner air.	Wind can disperse pollution, and humidity is associated with lower combined pollution in this dataset.
Which cities have the highest risk of bad-air days?	The global top-10% bad-air threshold is about 0.267 NPI. Dubai dominates bad-air days, followed by Tokyo, Nairobi, Cairo, and New York.	Dubai's profile combines high PM2.5, O3, NO2, temperature, and low green space, creating persistent risk.

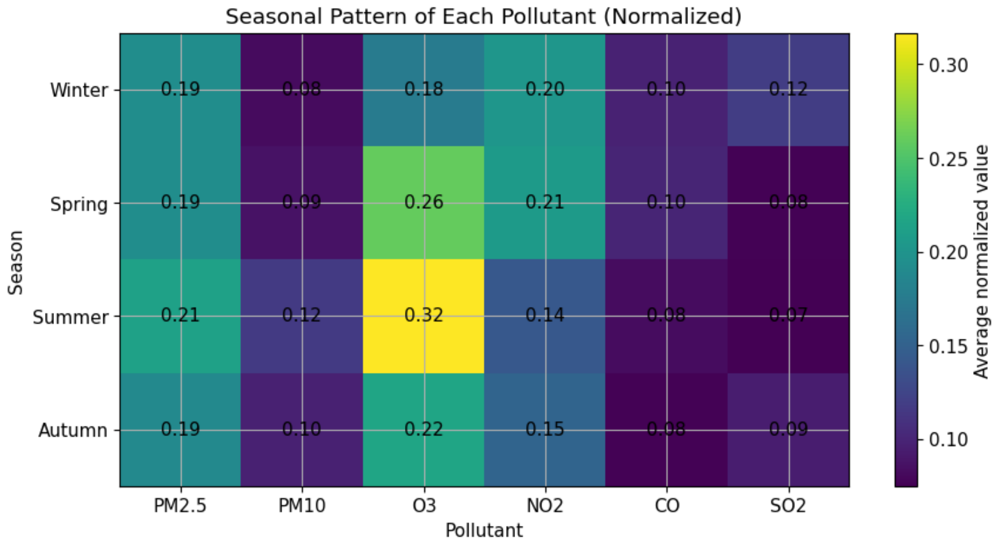


Figure 9. Seasonal pollution evidence for the extra questions.

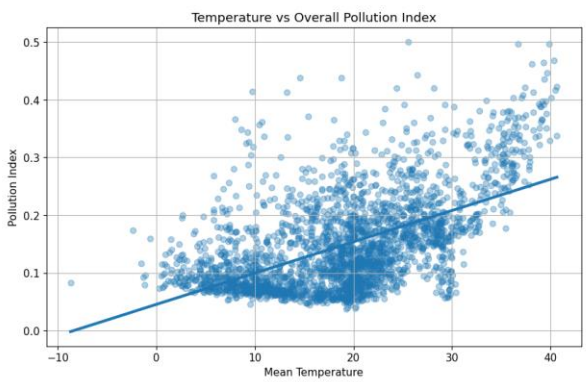


Figure 10. Weather and pollution evidence.

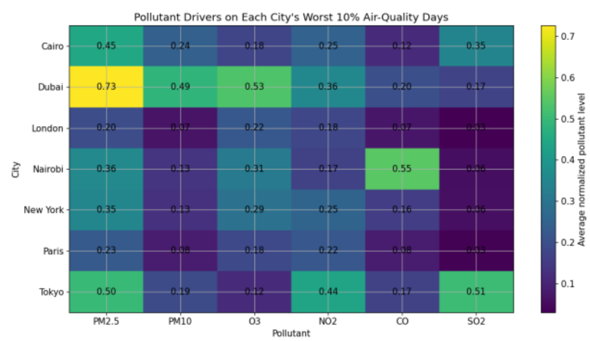


Figure 11. Bad-air-day driver evidence.

Outlier Analysis

The outlier analysis shows that extreme observations are concentrated mainly in pollution-related features rather than stable urban features such as green space. Across all cities, outliers occur more often on weekdays than weekends, strengthening the interpretation that human activity and workday movement contribute to unstable air-quality episodes.

City	Outlier Pattern
Cairo	Humidity and particulate pollution show high variability. Winter has the most outliers, and low wind conditions appear to allow pollution accumulation.
Dubai	NO2, O3, PM10, and PM2.5 drive extreme events, especially in summer and spring. Humidity outliers tend to be lower than normal.
London	Winter has the most outliers. NO2 and humidity increase during abnormal periods while several other weather variables decline.
Nairobi	NO2, CO, PM2.5, and PM10 outliers indicate episodic pollution. High CO events are especially strong and often occur with lower wind speed.
New York	Wind speed and CO are important outlier drivers. Summer and winter show stronger variability than spring.
Paris	CO and SO2 outliers dominate, with winter showing the largest abnormality count.

Tokyo	SO2, NO2, O3, PM10, and PM2.5 drive outliers. Winter shows the most variability.
Electricity	New York shows a July electricity-consumption outlier, consistent with peak summer cooling demand.

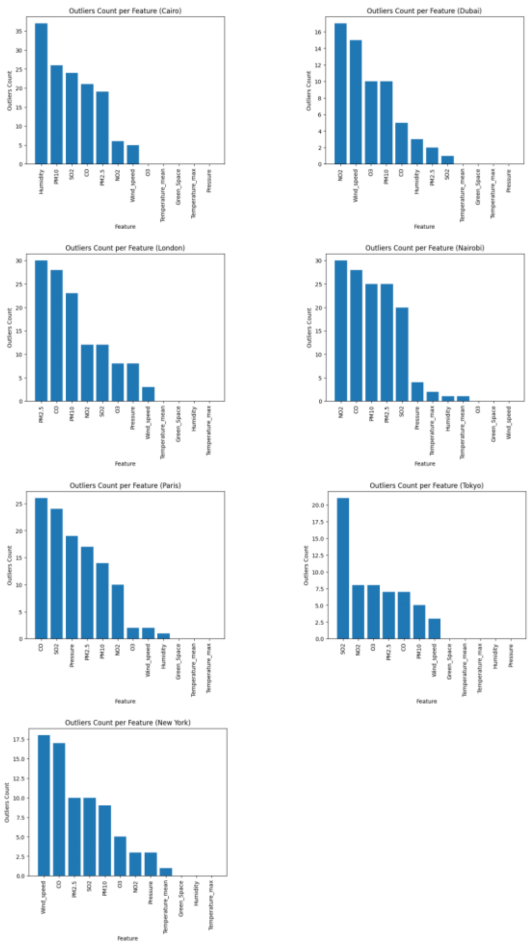


Figure 12. Outlier feature patterns.

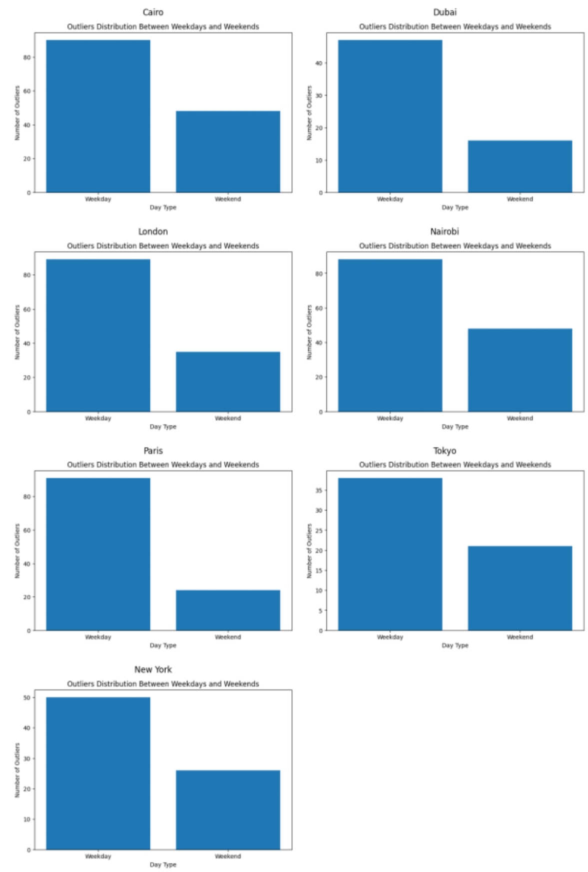


Figure 13. Weekday/weekend outlier comparison.

PM2.5 Predictive Modeling

The supervised modeling target is PM2.5. The selected model is a Random Forest Regressor wrapped in a preprocessing pipeline. Numerical features are standardized, while City and Season are one-hot encoded. The model uses PM10, NO2, CO, SO2, Green_Space, Temperature_mean, Humidity, Wind_speed, City, and Season as inputs.

Model Component	Documentation
Algorithm	Random Forest Regressor
Best parameters	n_estimators=100, max_depth=10, min_samples_leaf=2, min_samples_split=10
Validation R2	0.9085
Train R2	0.9912
Overfitting gap	0.0827, indicating strong performance with moderate overfitting risk
Strongest correlations with PM2.5	PM10 (0.900), NO2 (0.585), SO2 (0.573), Temperature_mean (0.533), CO (0.503); negative with Green_Space (negative 0.558) and Humidity (negative 0.393)

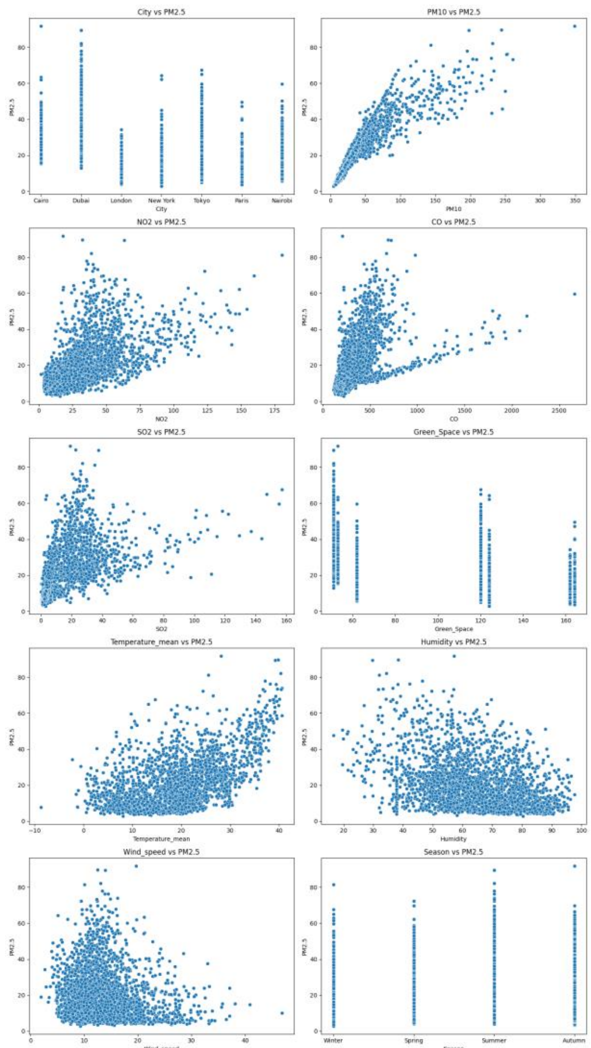


Figure 14. PM2.5 relationship visualization.

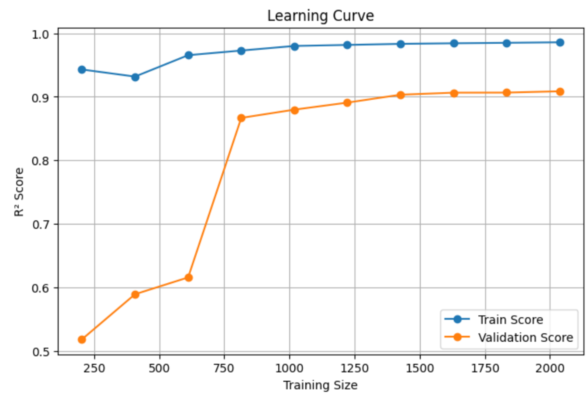


Figure 15. PM2.5 model learning curve.

The feature-importance chart shows that PM10 dominates the supervised PM2.5 model. This makes practical sense because PM10 and PM2.5 are closely related particulate measures, but it also means the model partially relies on a near-proxy feature. For application use, this is acceptable when PM10 is available, but future work could test a model that excludes PM10 to evaluate prediction under lower-sensor conditions.

Application Layer

The Streamlit PM2.5 dashboard allows a user to select a city and season, enter pollution indicators and weather/environmental values, and receive a predicted PM2.5 concentration. The app translates the predicted value into health categories such as Good, Moderate, Unhealthy for Sensitive Groups, Unhealthy, and Hazardous. It also displays a gauge, pollutant comparison chart, and recommendations based on the predicted risk level.

A second Streamlit page loads the humidity cluster model. The user inputs PM2.5, O3, NO2, CO, SO2, temperature, and wind speed. The app returns the assigned cluster, predicted humidity, confidence, and lower/upper uncertainty bounds.

Cluster Modeling and Overall Cluster Analysis

Cluster analysis was used to discover environmental structures that are not obvious from single charts. The project includes three clustering views: city-level clustering, seasonal city clustering, and predictive humidity clustering. Together they show that cities group more by environmental profile than by geography alone.

City-Level Clustering

For city clustering, each city was summarized by whole-year environmental profiles. StandardScaler was applied before distance-based clustering so electricity, pollutants, humidity, and weather variables would contribute fairly. Redundant features such as PM10, pressure, and temperature maximum were removed in the clustering logic because they either duplicated another signal or reflected altitude rather than environmental quality.

Cluster View	Cities / Result	Insight
Hot arid / high pollution	Cairo and Dubai in the K-Means interpretation	Highest PM2.5, warmer temperatures, lower green space, and lower humidity create a clear hot urban profile.
Clean temperate / moderate	London, Nairobi, New York, Paris	Lower PM2.5 and SO2 with stronger green-space or humidity signals. Nairobi joins this group in K-Means despite its different geography, showing that cluster similarity is data-driven.
Tokyo singleton	Tokyo	Tokyo separates because of high electricity consumption, high wind speed, strong NO2, and distinctive SO2 peak behavior.
Hierarchical nuance	Dubai and Tokyo appear as unusually distinct in the hierarchical dendrogram	The disagreement between hierarchical clustering and K-Means is informative: Dubai sits near the boundary between Cairo-like hot pollution and singleton extremity.

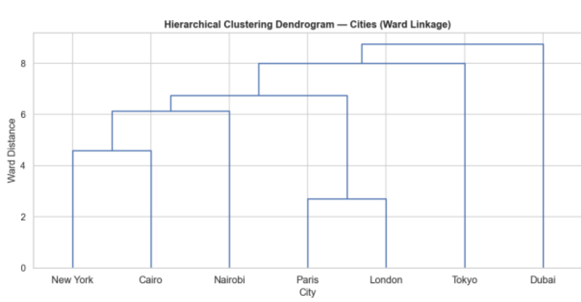


Figure 16. City-level clustering dendrogram.

Figure 17. City cluster profiles.

Overall city-clustering insight: climate zone, environmental structure, and development profile explain the data better than city size or continent alone.

Seasonal Clustering

Seasonal clustering aggregates the data into 28 city-season observations. The selected solution uses k=4 and shows perfect agreement between hierarchical clustering and K-Means after label alignment (Adjusted Rand Index = 1.0000). The silhouette score is 0.3316 and the Davies-Bouldin index is 1.0182, which is reasonable for real-world environmental data with only 28 observations.

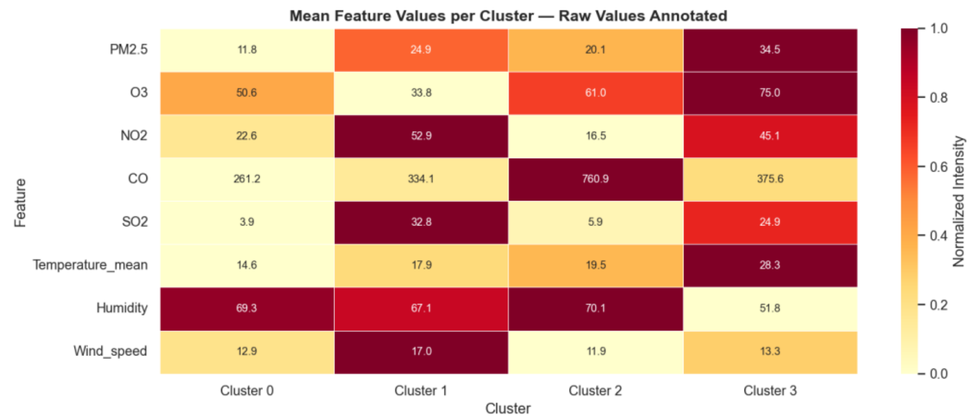


Figure 18. Seasonal cluster profiles.

Seasonal Cluster	Profile	Overall Interpretation
Cluster 0	Clean temperate	London, Paris, New York, and some Nairobi seasons remain low-pollution and humid compared with the rest of the dataset.
Cluster 1	High NO2/SO2, windy	Tokyo seasons and Cairo winter share stronger NO2/SO2 behavior, showing a cross-city pattern that calendar seasons alone do not explain.
Cluster 2	Equatorial CO pattern	Nairobi spring and summer form their own high-CO, humid cluster.
Cluster 3	Hot polluted	Dubai all year plus most Cairo seasons form a warm, lower-humidity, higher-pollution cluster.

Overall seasonal-clustering insight: clusters are city-driven more than season-driven. Dubai shows almost no seasonal movement, London/Paris/New York remain structurally stable, Nairobi splits by rainy versus drier periods, and Cairo winter behaves differently from Cairo's other seasons.

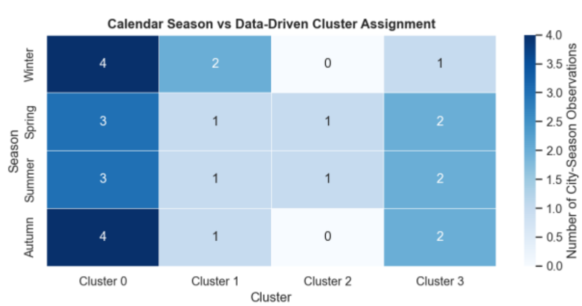


Figure 19. Season vs cluster distribution.

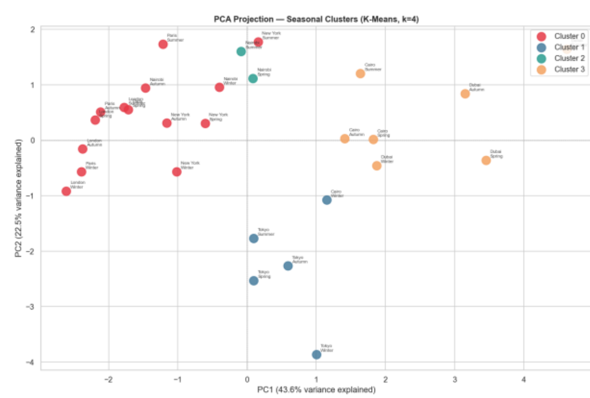


Figure 20. PCA projection.

Predictive Humidity Clustering

The humidity model uses K-Means as a prediction mechanism rather than a traditional regression model. Humidity is hidden during clustering. The model groups records using PM2.5, O3, NO2, CO, SO2, Temperature_mean, and Wind_speed. Each cluster's mean humidity becomes the predicted humidity for new records assigned to that cluster.

Item	Result / Documentation
Selected clusters	k=2, selected by the best silhouette score
Prediction outputs	Cluster ID, predicted humidity, confidence, lower bound, upper bound
Cluster humidity means	Cluster 0: 57.47%; Cluster 1: 68.31%
Evaluation	MAE 11.52%, RMSE 13.85%, R2 0.1307; baseline mean MAE 12.22% and RMSE 14.86%
Interpretation	The model is an exploratory approximation. It performs better than a naive mean baseline, but its predictions should be understood as cluster-level estimates rather than exact humidity forecasts.

Figure 21. Predictive humidity clustering k-selection.

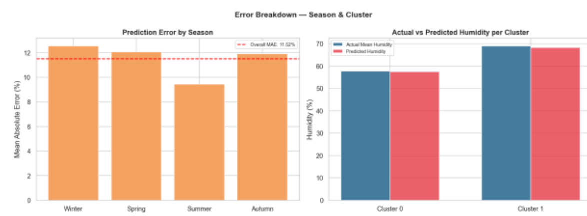


Figure 22. Predictive humidity clustering error analysis.

Overall cluster-analysis conclusion: clustering revealed meaningful latent environmental structures within the dataset. Environmental and atmospheric variables contained distinguishable patterns that support environmental monitoring and exploratory prediction tasks.

Limitations

The electricity feature is monthly and expanded to daily records, so it should not be interpreted as true daily electricity demand. Green space is static and assumed stable for 2023. Weather coverage differs slightly by city, and New York required backfilling for missing temperature and humidity values. The PM2.5 model performs very well, but its strongest feature is PM10, which is closely related to the target. Finally, clustering results are exploratory because the city-level sample size is small.

Final Conclusion

This project produced a clean, merged, multi-source urban environmental dataset and used it to answer sustainability questions from several angles. The strongest finding is that greener and more temperate city profiles tend to show cleaner air, while hotter and lower-green-space profiles show higher pollution pressure. Weekends are somewhat cleaner than weekdays, pollution is worse in summer overall, and weather conditions such as heat, dryness, and low wind contribute to worse air-quality conditions.

The modeling layer turns the analysis into a usable system. The PM2.5 Random Forest model supports user-facing prediction with strong validation performance, while humidity clustering demonstrates an interpretable cluster-based estimation approach. Environmental similarity is not just geographic: cities and seasons form meaningful groups based on pollution and atmospheric behavior.

For future work, the project could add true daily electricity data for every city, test PM2.5 prediction without PM10, expand to more cities, and add external variables such as traffic, population density, industrial activity, and policy indicators.

Source Files Used

Project Area	Files Reviewed
Narrative and planning	README.md, plan.txt, Features.txt, links.txt
Raw and integrated data	Data/raw/*.csv, Data/integrated/*.csv, Data/final/environment_data.csv
Collection code	src/scraping/*.py
Merging and preprocessing	merge.ipynb, handle_concatenate_weather.ipynb, preprocessing.ipynb
EDA and insights	green_space_air_quality_analysis.ipynb, daytype_air_quality.ipynb, Temperature_vs_Electricity1.ipynb, abdelrahman_environmental_questions.ipynb, Outliers_visualizations.ipynb, outlier_*.ipynb
Modeling and clustering	src/Modeling/*.ipynb, src/Modeling/humidity_model.py, src/Clustering/*.ipynb, Models/*.pkl
Applications	src/Application/App.py, src/Application/Humidity_model_App.py